



We design frontier systems that work in production.

We get things done, excellently.

CONFIDENTIALITY NOTICE

This portfolio is prepared solely for showcasing selected works, capabilities, and professional experience of the Kalana Consulting and its team. All materials, including project references, visuals, concepts, methodologies, technical documentation, and case examples contained in this document are intended for portfolio review purposes only.

No part of this document may be copied, reproduced, distributed, modified, published, or used for commercial, operational, or derivative purposes without prior written consent from the rightful owner. Any unauthorized use, disclosure, or duplication of this document, whether in whole or in part, is strictly prohibited.



BPJS Kesehatan
Badan Penyelenggara Jaminan Sosial



AI Incubation
For Public Sector

TEAM KALANA

ClaimSight AI

**AI-POWERED HEALTH CLAIMS FRAUD
DETECTION
FOR BPJS KESEHATAN**

AI for Public Sector

Claims Fraud: A Hidden Threat Draining Indonesia's National Health Fund

IDR 20 T

Estimated annual fraud loss
in JKN program (KPK, 2024)

IDR 866 B

Fraud potential detected
by BPJS (verification Nov 2023)

129 K+

Fraud cases detected
in 2024 (BPK Audit, 2026)

4,341

Fictitious claims found in
3 sampled hospitals (KPK, 2024)

Root Cause: Why Has This Not Been Solved?

Verification is still manual & sample-based

BPJS relies on sampling-based verification, not systematic detection. Fraud outside the sample goes undetected.

KONAN FIRST monitoring system underperforms

BPK (2026): the internal oversight system is inaccurate and incomplete. Internal audit (SPI) fraud reports are not submitted regularly.

Fraud patterns continuously evolve

2023: phantom billing (40% of cases). 2024: hemophilia therapy manipulation (52%). 2025: ICU ventilator manipulation. Static rules cannot keep up.

Other Countries Have Succeeded — Indonesia Can Do Better

USA – CMS Medicare	China – National Medical Insurance	Global Baseline (WHO / GHAN)
WHAT THEY DID: AI + ML deployed to detect fraud across 4.5 million claims/day. Fraud Detection Operations Center analyzes claims in real-time before payment. ✓ RESULTS: Saved \$2 billion (Mar 2025–Feb 2026). 90% of bad actors identified within the first year of AI deployment. ⚠ LIMITATIONS: System relies on public Medicare data; difficult to replicate in closed systems. <i>Source: nextgov.com/artificial-intelligence/2026/02/cms-saved-2-billion-using-ai-fight-fraud/411661/</i>	WHAT THEY DID: AI and graph analysis detect networks of providers involved in collective fraud. \$339 million USD recovered from health insurance fraud in 2020 alone. ✓ RESULTS: Network-based fraud detection is more effective at catching organized fraud rings (doctor–hospital collusion). ⚠ LIMITATIONS: Different data context; China has a more integrated mandatory identity system. <i>Source: ncbi.nlm.nih.gov/pmc/articles/PMC10080767/</i>	WHAT THEY DID: An estimated 3–10% of global healthcare spending is lost to fraud. Manual detection: inconsistent, dependent on whistleblowers. ✓ RESULTS: ML-based detection proven to improve accuracy & consistency — catching patterns invisible to human auditors. ⚠ LIMITATIONS: Labeled fraud data is very scarce in developing countries — semi-supervised approaches required. <i>Source: sciencedirect.com/science/article/pii/S0933365724003038</i>

Insight: ClaimSight adopts an ensemble AI approach (like CMS) + graph network analysis (like China), customized for Indonesia's INA-CBGs data context and BPJS SOPs.

ClaimSight AI — Three Layers of Detection

A multi-layer AI system that detects anomalies, classifies risk, and prioritizes audits — without manually reviewing every single claim.

01

Anomaly Detection (Unsupervised)

Model: Isolation Forest + DBSCAN

Detects unusual patterns without requiring pre-labeled fraud data. Effective against new, previously unseen fraud schemes.

- Sudden spike in patient visit counts
- Unusual procedures for a given diagnosis
- Claims filed outside hospital operating hours/days

02

Risk Classification (Supervised)

Model: XGBoost + Random Forest Ensemble

Trains on historically labeled fraud data.
Output: a risk score from 0–100 per claim.

- Upcoding (billing under a costlier diagnosis code)
- Phantom billing (claiming services never rendered)
- Procedure manipulation (ventilator → nasal cannula)

03

Network Analysis (Graph AI)

Model: Graph Neural Network (GNN)

Uncovers organized fraud rings — doctor–hospital–fictitious patient collusion invisible when inspecting claims individually.

- Referral loops within a single provider network
- Collective billing patterns across affiliated hospitals
- Identifying the 'ringleader' of fraud consortiums

OUTPUT: Periodic risk reports + real-time dashboard → auditors only review high-risk claims, not the entire daily volume.

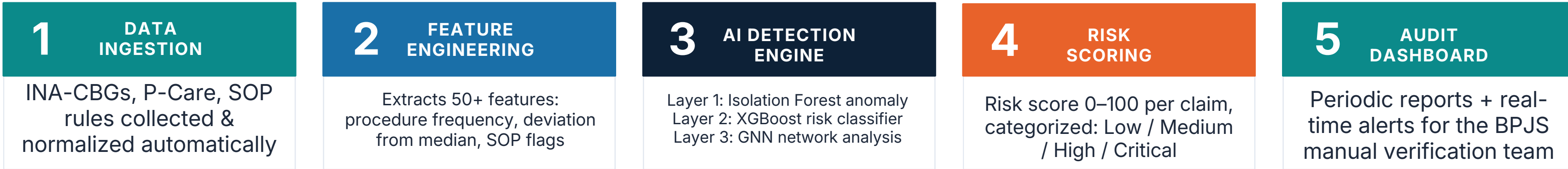
Data Needed: Available, Structured, Protected

DATA CATEGORY	SOURCE	TYPE	AVAILABILITY	SAMPLE FEATURES
Claims & Services	INA-CBGs BPJS Kesehatan	Structured	Available (BPJS internal)	Diagnosis codes, procedures, rates, dates, hospital
Provider Profiles	BPJS Credentialing System	Structured	Available (BPJS internal)	Doctor specialization, patient ratio, sanction history
Patient Medical Records	P-Care / i-Care JKN	Semi-structured	Available (restricted access)	Digital medical records, visit history
Historical Fraud Labels	PK-JKN Team / BPKP / KPK	Labeled	Partial – needs normalization	Verified cases 2020–2024 (109K+ cases)
ICD-10 / SOP Reference	Ministry of Health / PNPk	Reference	Publicly available	Diagnosis standards, procedures, INA-CBGs rates

DATA ACCESS PLAN:

Incubation Phase: Anonymized data for model development via MoU with BPJS Kesehatan. **Piloting Phase:** Sandbox access (regulatory sandbox) with real data, supervised by the PK-JKN team. **Security:** End-to-end encryption, role-based access control, audit log on every query.

ClaimSight AI System Flow



Who Uses ClaimSight?

PRIMARY

**BPJS Kesehatan
— Verification & Anti-Fraud Control
Team (PK-JKN)**

Current Pain Points:
Must verify hundreds of thousands of claims/month. Limited team capacity. Fraud schemes evolve rapidly.

What They Gain:

- Real-time risk score dashboard
- Prioritized claim queue for manual audit
- Automated alerts for Critical Risk claims
- Periodic reports per hospital/doctor
- Explainable AI: why a claim was flagged

SECONDARY

KPK & BPKP Enforcement Division

Receives consolidated risk reports for further investigation. Can request deep-dive analysis on specific providers.

SECONDARY

Ministry of Health

Monitors national fraud pattern trends. Provides inputs for Permenkes updates and PNPK clinical standards.

TERTIARY

Hospitals (Self-Check)

Hospitals can use an open-access version of ClaimSight to pre-check their own claims before submission to BPJS.

Targeted & Measurable Outcomes



QUALITATIVE IMPACT

-  **Public trust in the JKN program increases** — funds reach the right beneficiaries
-  **BPJS Kesehatan can narrow its operational deficit** (IDR 7.9 T in 2023)
-  Model continuously learns from audit results → **more accurate** every month (continuous learning)
-  **Replication potential across 34 provinces** & other social insurance systems (BPJS Ketenagakerjaan)

8-Week Incubation Roadmap

PERIOD	PHASE	KEY DELIVERABLE
Week 1–2	Onboarding & Data	✓ Sandbox access granted, INA-CBGs data normalized, initial exploratory data analysis
Week 3–4	Feature Engineering & Model v1	✓ 50+ features engineered; baseline Isolation Forest + XGBoost trained
Week 5–6	Piloting & Iteration	✓ Model tested on real BPJS data (anonymized); GNN network analysis layer added
Week 7	Dashboard & Integration	✓ Risk-scoring dashboard built; API integration with BPJS verification system
Week 8	Demo Day & Report	✓ Presentation to BPJS + KPK; accuracy, precision, recall report; scale-up plan submitted

Key Risks & Mitigation Strategies

LEVEL	RISK & IMPACT	MITIGATION STRATEGY
HIGH	Limited labeled fraud data <i>Impact: Supervised model accuracy is low without quality training data</i>	Mitigation: Semi-supervised approach: combine unsupervised (Isolation Forest) with incremental supervised learning. Coordinate fraud labels with PK-JKN team & BPKP.
HIGH	Resistance from hospitals/providers <i>Impact: Hospitals may refuse cooperation or submit claims in formats that obstruct analysis</i>	Mitigation: Frame as 'quality assurance', not 'surveillance'. Engage PERSI (hospital association) from day one. ClaimSight output also usable by hospitals for self-check.
MEDIUM	High false positives → verifier overload <i>Impact: Audit team overwhelmed if too many claims are incorrectly flagged</i>	Mitigation: Set high precision threshold (>85%). Display per-claim explainability (SHAP values) so verifiers can make fast, informed decisions.
MEDIUM	Privacy & security of JKN member data <i>Impact: Risk of medical data breach for millions of members</i>	Mitigation: Data anonymization from ingestion. AES-256 encryption. Role-based access control. Compliance with Personal Data Protection Law No.27/2022.
LOW	Fraud pattern drift (schemes change fast) <i>Impact: A model trained today may miss new fraud schemes next month</i>	Mitigation: Continuous learning: monthly re-training with new audit data. Feature distribution monitoring (data drift detection via Evidently AI).

TEAM KALANA



Salma Kurnia Dewi

Product Manager



**Patrick Samuel Evans
Simanjuntak**

Tech Lead



Mariano Gerardus Senduk

Engineer